

def 1.0. Define the set of integers as $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$.

def 2.1. An integer n is even if $n = 2k$ for some $k \in \mathbb{Z}$.

def 2.2. An integer n is odd if $n = 2k + 1$ for some $k \in \mathbb{Z}$.

Fact 2.3.1. Every integer is either even or odd, but never both.

Fact 2.3.2 Sums and products of integers are integers.

def 2.11. An integer n divides the integer m written $\frac{m}{n}$, if and only if, there exists $k \in \mathbb{Z} : m = nk$.

Theorem 2.4.

Claim: If $z \in \mathbb{Z}$ then the sum of z and $z + 1$ is odd.

Proof:

Assume that $z \in \mathbb{Z}$.

Note that $1 \in \mathbb{Z}$, making it an integer by *def 1.0*.

Since both 1 and z are integers, $z + 1 \in \mathbb{Z}$ by *Fact 2.3.2*.

Once again, we know $z + (z + 1)$ must be an integer by *Fact 2.3.2*.

Now $z + z + 1 = 2z + 1$.

Since $z \in \mathbb{Z}$, let $k = z$ by *def 2.2*.

So $2z + 1 = 2k + 1$

Therefore, by *def 2.2*, the sum of z and $z + 1$ is odd.



Problem 2.9.

Claim: The product of two odd integers is odd.

Proof:

Assume x and y are odd integers.

Using *def 2.2*, fix values of k called k_x, k_y so that $x = 2k_x + 1, y = 2k_y + 1$.

Then, $x * y = (2k_x + 1)(2k_y + 1) = 4k_x k_y + 2k_x + 2k_y + 1 = 2(2k_x k_y + k_x + k_y) + 1$.

Note that 2 is an integer by *def 1.0* and k_x, k_y are integers because of *def 2.2*.

Thus, by *Fact 2.3.2*, the term $2k_x k_y + k_x + k_y \in \mathbb{Z}$.

So, by *def 2.2*, let $k = 2k_x k_y + k_x + k_y$.

Then $x * y = 2(2k_x k_y + k_x + k_y) + 1 = 2k + 1$.

Thus, by *def 2.2*, the product of two odd integers is odd.



Theorem 2.17.

Claim: If $n, m \in \mathbb{Z}$ and n divides m , then n divides $-m$.

Proof:

Assume that $n, m \in \mathbb{Z}$, since n divides m , using *def 2.11*, fix an integer value of k called k' so that $m = nk'$. Note that -1 is an integer by *def 1.0*. We can multiply both sides of the equation by the integer -1 and it will remain equal. Doing so, $-m = n(-k')$. Since $-k'$ is an integer by *fact 2.3.2*, we can let $k = -k'$, so $-m = nk$. Thus, by *def 2.11*, n divides $-m$. Therefore, if $n, m \in \mathbb{Z}$ and n divides m , then n also divides $-m$.



Theorem 2.21.

Claim: If $n, p, m \in \mathbb{Z}$ are such that n divides p and p divides m , then n divides m .

Proof:

Assume that $n, p, m \in \mathbb{Z}$ and n divides p and p divides m . Using *def 2.11* fix integer values of k called k_p, k_m so that $p = nk_p, m = pk_m$. Note that k_p and k_m are integers, by *fact 2.3.2*, their product is also an integer. With, *def 2.11*, let $k = k_p k_m$. Now,

$$m = pk_m = nk_p k_m = nk.$$

Thus, $m = nk$, and by *def 2.11*, this means n divides m . Therefore, if $n, p, m \in \mathbb{Z}$ and n divides p and p divides m , then n also divides m .

